

Basic Electrical Engineering

MODULE 1 — PART 1: COMPLETE NOTES

Module 1 (10 hrs): DC Circuits

For First-Year Students — written in simple, easy language

Course: Basic Electrical Engineering · Course Code: EE100

Module 1 — what it covers (as per the syllabus)

- Ohm's law and Kirchhoff's laws
- Analysis of series, parallel and series-parallel circuits excited by independent voltage sources
- Mesh analysis and Nodal analysis
- Star-Delta and Delta-Star conversion
- Superposition theorem
- Thevenin's and Norton's theorems
- Maximum power transfer theorem

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1. Basic Circuit Elements and Sources

1.1 What is an Electric Circuit?

An **electric circuit** is an interconnection of various electrical elements in which there is at least one closed path in which current can flow. Circuit elements are of two kinds:

- **Active elements** — capable of supplying energy to the circuit. Examples: voltage sources, current sources and generators (alternators, DC generators, PV cells).
- **Passive elements** — can only receive energy, which they either dissipate, absorb or store. Examples: resistors, inductors and capacitors.

1.2 Voltage, Current, Power and Energy

Voltage

Voltage (or potential difference) is the energy required to move a unit charge through an element, measured in volts (V). The voltage between two points a and b is the work needed to move a unit charge from a to b, and in a circuit it is often defined relative to ground.

$$v = dW / dq \quad (W = \text{work/energy in joules, } q = \text{charge in coulombs})$$

Current

Electric current is the time rate of change of charge, measured in amperes (A).

$$i = dq / dt$$

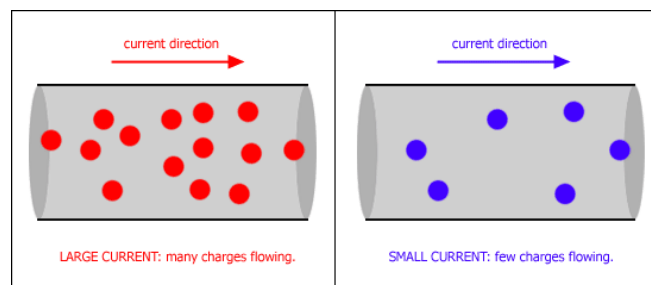


Figure 1: Electric current as the flow of charge.

Power

Power is the rate of change of energy, measured in watts (W).

$$p = dw / dt = v \cdot i = i^2 R = v^2 / R$$

- If power is **positive**, power is being delivered to (absorbed by) the element.
- If power is **negative**, power is being supplied by the element.
- **Passive sign convention:** it is satisfied when the current enters the positive terminal of an element, giving $p = +vi$; if the current enters the negative terminal, $p = -vi$.

Energy

Energy is the capacity to do work, measured in joules (J). Electrical energy = Power × Time. Utility companies measure energy in watt-hours (Wh); 1 Wh = 3600 J.

1.3 Types of Sources

Sources are classified as independent or dependent:

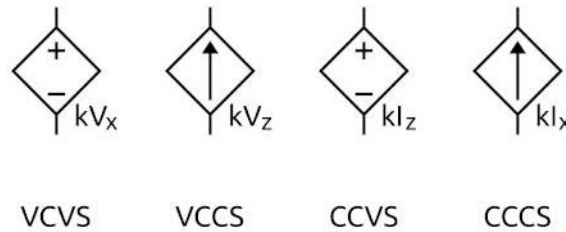


Figure 2: Symbols of the ideal voltage and current sources.

Independent Sources

An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit variables.

- An **ideal voltage source** maintains a constant terminal voltage no matter how much current is drawn from it.
- A **practical voltage source** has a low but finite internal resistance R_{int} , which causes its terminal voltage to decrease as the load current increases (and vice-versa).
- An **ideal current source** supplies the same current to any load resistance connected across its terminals.
- A **practical current source** has a high but finite internal resistance; therefore the load current changes as the value of the load resistance changes.

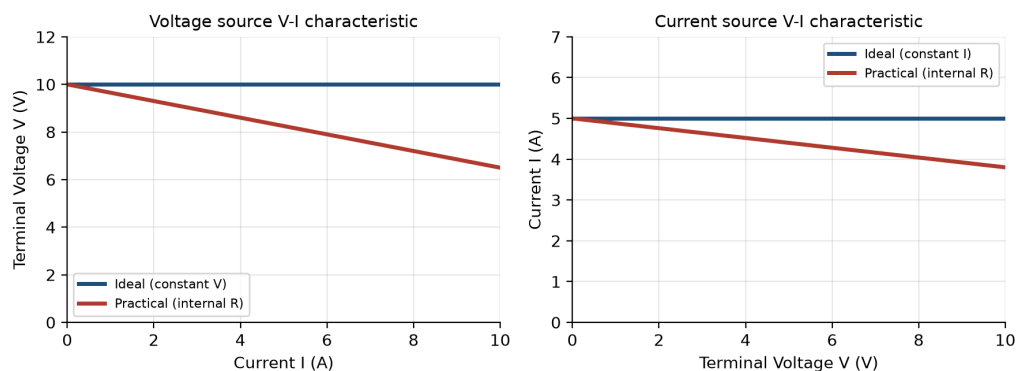


Figure 3: V-I characteristics of ideal and practical voltage and current sources.

Dependent (Controlled) Sources

Special sources in which the source voltage or current depends on some other quantity in the circuit (a voltage or a current anywhere in the circuit) are called **dependent** or controlled sources. They are of four types:

- Voltage-Controlled Voltage Source (VCVS)
- Current-Controlled Voltage Source (CCVS)
- Voltage-Controlled Current Source (VCCS)
- Current-Controlled Current Source (CCCS)

1.4 Resistor

Resistance is that property of a circuit element which opposes the flow of electric current and in doing so converts electrical energy into heat energy. It is the proportionality factor in Ohm's law relating voltage and current. The unit of resistance is the **ohm** (Ω); one ohm is one volt per ampere.

$$V \propto I \Rightarrow V = IR$$

$$\text{Power drawn by a resistor: } P = VI = I^2R = V^2/R$$

The resistance of an electrical conductor depends on four factors:

- 1 the length of the conductor (l)
- 2 the cross-sectional area of the conductor (a)
- 3 the type of material (resistivity ρ)
- 4 the temperature of the material

$$R = \rho l / a \quad (\text{resistivity } \rho \text{ is measured in ohm-metre, } \Omega\text{-m})$$

1.5 Inductor

If a time-varying current flows through a coil, an emf is induced in it. The induced emf across the coil is directly proportional to the rate of change of current with respect to time. Because of this property of inducing emf, all types of electrical coil can be referred to as **inductors**. An inductor is an **energy storage device** that stores energy in the form of a magnetic field, and it opposes any change in the amount of current flowing through it.

$$v_L = L \cdot di/dt$$

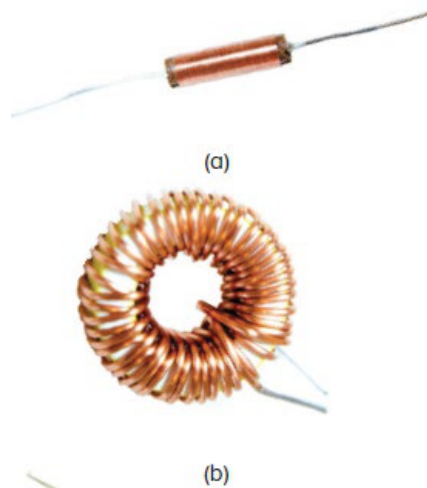


Figure 4: Inductor symbols.

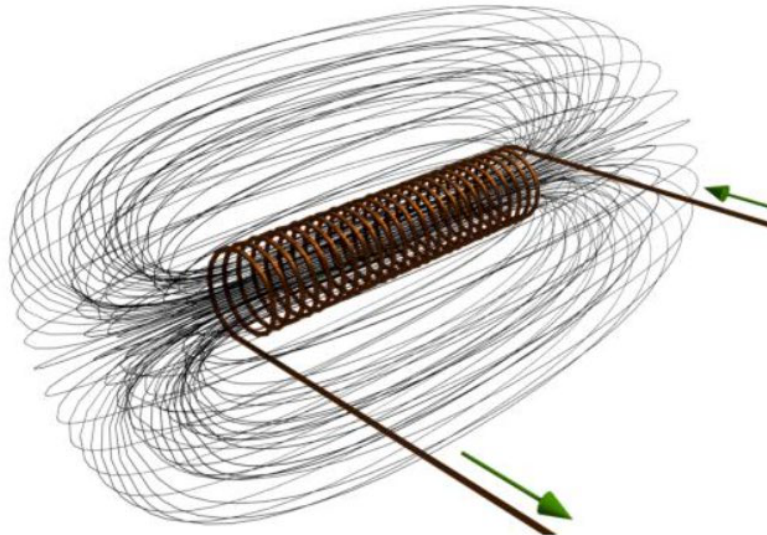


Figure 5: Magnetic field lines around an inductor carrying current.

The inductance of an electrical conductor depends on four major factors: it is directly proportional to the square of the number of turns, directly proportional to the area of cross-section, inversely proportional to the length, and depends on the absolute permeability of the magnetic material:

$$L = \mu N^2 A / l$$

1.6 Capacitor

A **capacitor** behaves as a charge-storage device. It holds electric charge when a voltage is applied across it, and gives up the stored charge to the circuit when required. Its most basic construction consists of two parallel conductors (usually metallic plates) separated by a dielectric material. When a voltage source is connected across the capacitor, the plate attached to the positive terminal becomes positively charged and the plate connected to the negative terminal becomes negatively charged. Because of the dielectric between the conductors, ideally no charge can migrate from one plate to the other.

$$i_C = C \cdot dv/dt$$

1.7 Open Circuit and Short Circuit

- An **open circuit** is a circuit element with resistance approaching **infinity** — no current flows through it.
- A **short circuit** is a circuit element with resistance approaching **zero** — the full current flows with no voltage drop across it.

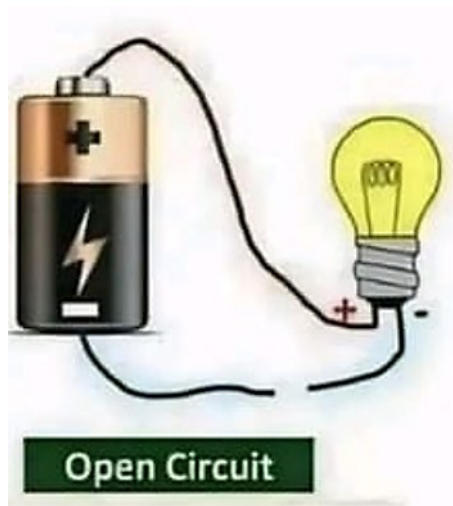


Figure 6: Open circuit.

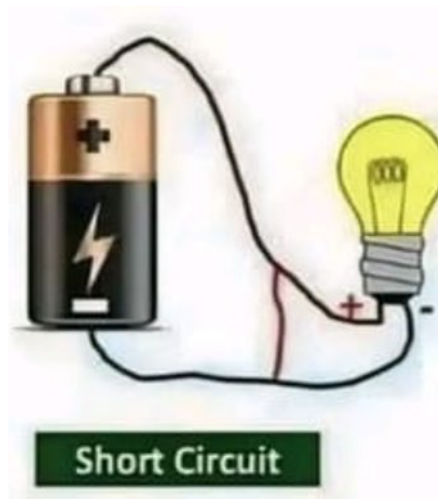


Figure 7: Short circuit.

2. Ohm's Law and Resistance Combinations

2.1 Ohm's Law

Ohm's law states that the current flowing through a conductor is directly proportional to the potential difference across it, provided the physical conditions (such as temperature) remain constant.

$$V = IR \quad I = V / R \quad R = V / I$$

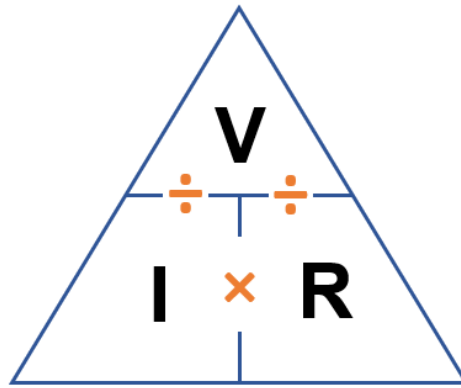


Figure 8: The Ohm's law triangle — $V = IR$.

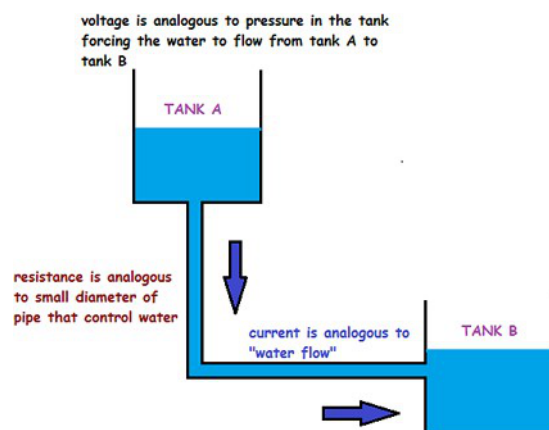


Figure 9: The water-flow analogy of voltage, current and resistance.

2.2 Series Circuits

Two or more elements are in **series** if they are cascaded (connected one after another) and consequently carry the **same current**. When a number of resistances are connected in series, the total (equivalent) resistance is equal to the sum of the individual resistances:

$$R_{eq} = R_1 + R_2 + R_3$$

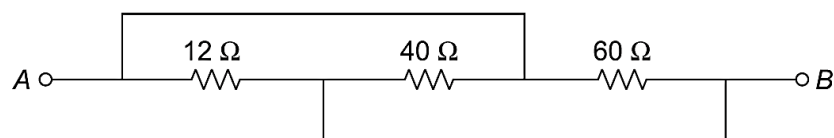


Figure 10: Three resistances in series.

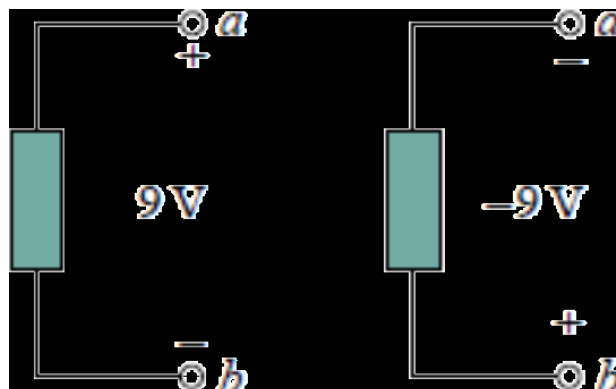


Figure 11: A series circuit with a voltage source.

Voltage division rule: in a series circuit the source voltage divides across the resistances in proportion to their resistance values. For two resistances:

$$V_1 = V \cdot R_1 / (R_1 + R_2) \quad V_2 = V \cdot R_2 / (R_1 + R_2)$$

2.3 Parallel Circuits

Two or more elements are in **parallel** if they are connected to the same two nodes and consequently have the **same voltage** across them. When a number of resistances are connected in parallel, the reciprocal of the total resistance is equal to the sum of the reciprocals of the individual resistances:

$$1 / R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$$

$$\text{For two resistances in parallel: } R_{eq} = R_1 R_2 / (R_1 + R_2)$$

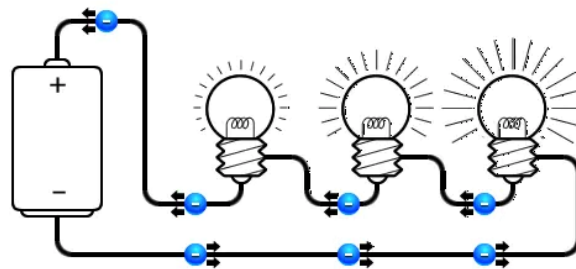


Figure 12: Two resistances in parallel.

Current division rule: in a parallel circuit the total current divides between the branches in inverse proportion to their resistances. For two resistances:

$$I_1 = I \cdot R_2 / (R_1 + R_2) \quad I_2 = I \cdot R_1 / (R_1 + R_2)$$

2.4 Series-Parallel Circuits

Most practical circuits are a combination of series and parallel parts. They are solved by reducing the circuit step by step — first replacing parallel groups by their equivalent resistance, then adding the series resistances, until a single equivalent resistance is obtained.

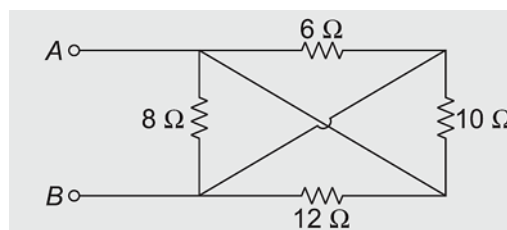


Figure 13: A series-parallel circuit.

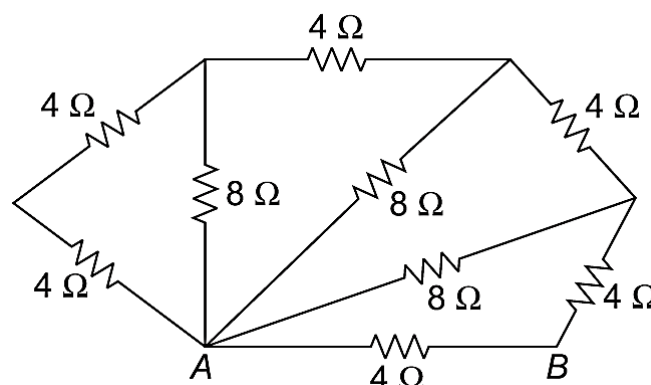


Figure 14: A series-parallel circuit.

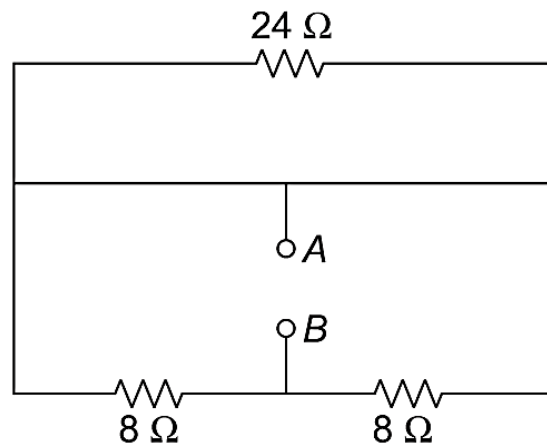


Figure 15: A series-parallel circuit.

3. Kirchhoff's Laws

3.1 Kirchhoff's Current Law (KCL)

KCL: the algebraic sum of the currents meeting at a junction in an electrical circuit is zero. In other words, the sum of the currents entering a node equals the sum of the currents leaving it.

$$I_1 + I_4 - I_2 - I_3 = 0 \Rightarrow I_1 + I_4 = I_2 + I_3$$

KCL is based on the law of conservation of charge — charge cannot pile up at a junction.

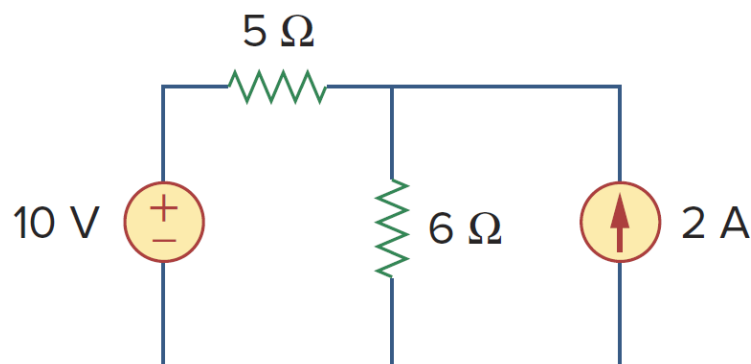


Figure 16: KCL at a junction — incoming currents equal outgoing currents.

3.2 Kirchhoff's Voltage Law (KVL)

KVL: in any closed electrical circuit or mesh, the algebraic sum of all the electromotive forces (emfs) and voltage drops in the resistors is equal to zero.

Algebraic sum of emfs + algebraic sum of voltage drops = 0

$$+V_1 - V_2 - V_3 + V_4 - V_5 = 0$$

KVL is based on the law of conservation of energy — the energy gained and lost around a closed loop must balance.

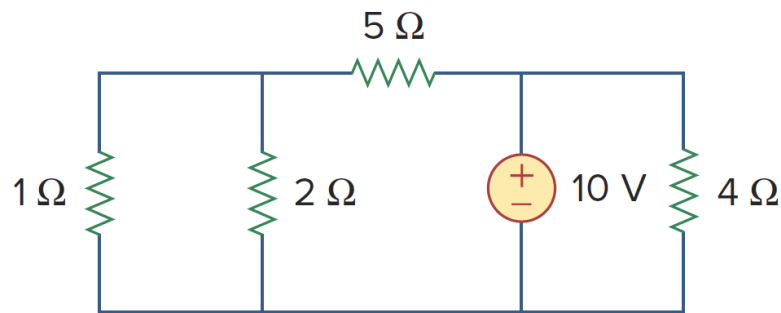


Figure 17: KVL around a closed loop.

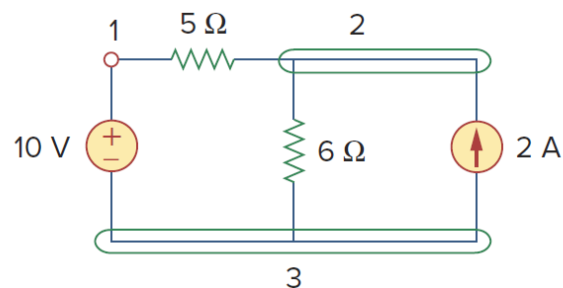


Figure 18: Applying Kirchhoff's laws to a circuit.

3.3 Circuit Topology — Nodes, Branches, Loops and Meshes

- **Node:** a point or junction where two or more of the circuit's elements meet.
- **Branch:** that part or section of a circuit which lies between two junctions.
- **Loop:** any closed path in a circuit.
- **Mesh:** a closed loop which contains no other loop within it (a path that does not contain any other path).

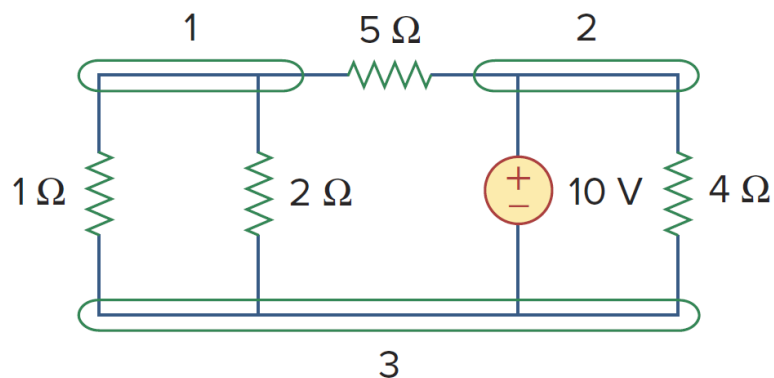


Figure 19: Nodes, branches, loops and meshes in a circuit.

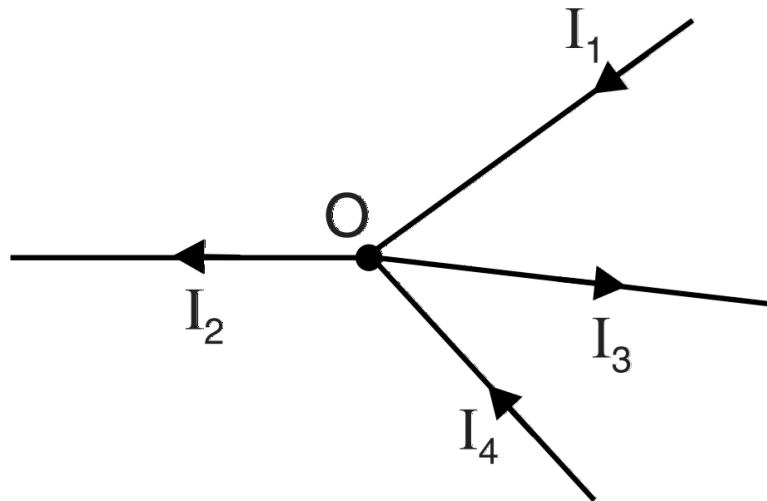


Figure 20: Loops and meshes marked on the same circuit.

4. Star-Delta and Delta-Star Conversion

A T-connected (or Y / star-connected) resistor network can be redrawn as an electrically equivalent star network, and a π -connected network can be converted into an equivalent Δ (delta) network. This conversion is very useful for simplifying networks that cannot be reduced by simple series and parallel combinations.

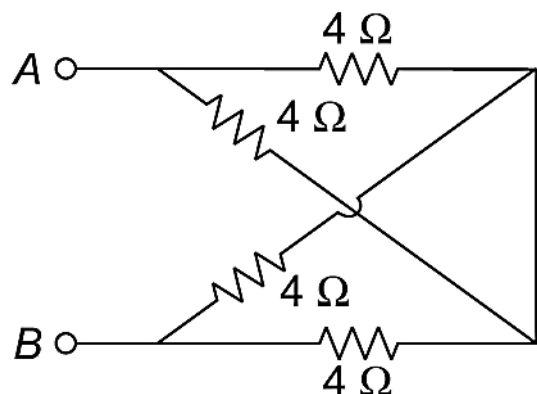


Figure 21: Star (Y) and delta (Δ) networks.

Delta to Star ($\Delta \rightarrow Y$)

Each star resistance equals the product of the two adjacent delta resistances divided by the sum of all three delta resistances:

$$R_1 = R_{AB} R_{CA} / (R_{AB} + R_{BC} + R_{CA}) \quad R_2 = R_{AB} R_{BC} / (\text{sum}) \quad R_3 = R_{BC} R_{CA} / (\text{sum})$$

Star to Delta ($Y \rightarrow \Delta$)

Each delta resistance equals the sum of the products of the star resistances taken in pairs, divided by the opposite star resistance:

$$R_{AB} = (R_1 R_2 + R_2 R_3 + R_3 R_1) / R_3$$

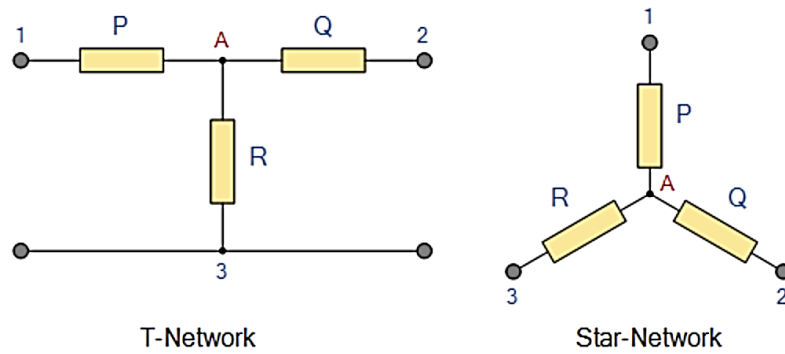


Figure 22: Delta network and its equivalent star network.

$$R_A = \frac{R_{AB} \cdot R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_B = \frac{R_{AB} \cdot R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_C = \frac{R_{BC} \cdot R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

Figure 23: Worked example — converting a delta network to an equivalent star network.

5. Mesh Analysis

A **mesh** is a loop which does not contain any other loops within it. Mesh analysis is applicable only to **planar** networks (a network is planar if it can be drawn on a plane surface without crossovers). In this method, a current is assigned to each mesh, and Kirchhoff's voltage law is written for each mesh in terms of the unknown mesh currents. If a network has a large number of voltage sources, mesh analysis is very useful.

5.1 Steps in Mesh Analysis

- 1 Identify the meshes, assign a direction (usually clockwise) and assign an unknown current to each mesh.
- 2 Assign the polarities for the voltage across the branches.
- 3 Apply KVL around each mesh and use Ohm's law to express the branch voltages in terms of the unknown mesh currents and the resistances.
- 4 Solve the simultaneous equations for the unknown mesh currents.

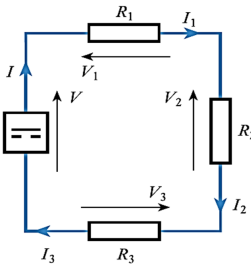
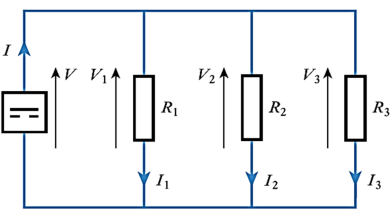
	Series circuit	Parallel network
		
Current	The current is the same in all parts of the circuit $I = I_1 = I_2 = I_3$	The total current supplied to the network equals the sum of the currents in the various branches $I = I_1 + I_2 + I_3$
Voltage	The total voltage equals the sum of the voltages across the different parts of the circuit $V = V_1 + V_2 + V_3$	The voltage across a parallel combination is the same as the voltage across each branch $V = V_1 = V_2 = V_3$
Resistance	The total resistance equals the sum of the separate resistances $R = R_1 + R_2 + R_3$	The reciprocal of the equivalent resistance equals the sum of the reciprocals of the branch resistances $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

Figure 24: A network with several meshes solved by mesh analysis.

For a network with meshes, the equations can be written in matrix form $[R][I] = [V]$, where the diagonal terms are the sum of the resistances of each mesh and the off-diagonal terms are the (negative) common resistances between meshes.

6. Nodal Analysis

In **nodal analysis**, one node is chosen as the reference (ground) node, and the voltages of all the other nodes are written with respect to this reference. Kirchhoff's current law is then applied at each non-reference node, expressing the branch currents in terms of the node voltages and resistances. This method is very useful when a network has a large number of current sources.

6.1 Steps in Nodal Analysis

- 1 Select one node as the reference (ground) node, and label the voltages of the other nodes $V_1, V_2, \dots$ with respect to it.
- 2 Assume all currents to move away from the node under consideration.
- 3 Apply KCL at each non-reference node, writing each branch current as (node voltage difference) / resistance.
- 4 Solve the simultaneous equations for the unknown node voltages.

For example, applying KCL at a node A connected to a node B through a resistance R , the current leaving A is $(V_A - V_B)/R$. The resulting equations are solved to give the node voltages, and any branch current can then be found using Ohm's law.

7. Source Transformation

- A voltage source with a series resistor can be converted into an equivalent current source with a parallel resistor.
- Conversely, a current source with a parallel resistor can be converted into a voltage source with a series resistor.

For equivalence: $I_S = V_S / R_S$ and $R_P = R_S$

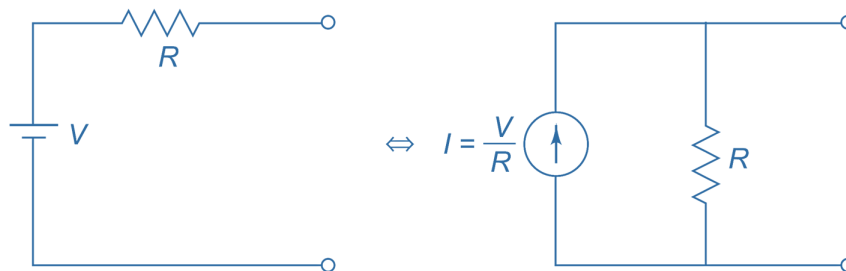


Figure 25: Voltage source with series resistance $\Leftrightarrow$ current source with parallel resistance.

8. Network Theorems

8.1 Superposition Theorem

In a linear, bilateral DC network containing more than one energy source, the resultant potential difference across (or current through) any element is equal to the algebraic sum of the potential differences (or currents) for that element produced by each source acting alone, with all other voltage sources replaced by short circuits and current sources replaced by open circuits.

Procedure

- 1 Select one source and replace all other voltage sources by short circuits and current sources by open circuits.
- 2 Determine the voltage across or current through the desired element due to this single source.
- 3 Repeat the above two steps for each of the remaining sources.
- 4 Algebraically add all the voltages or currents to obtain the actual value when all sources act together.

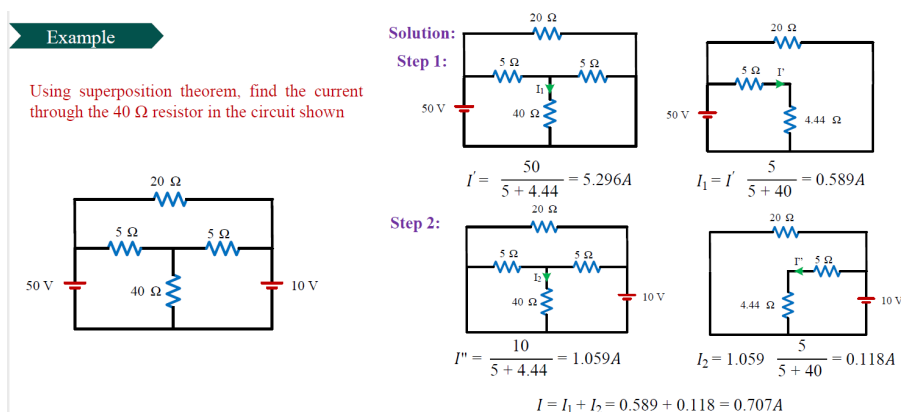
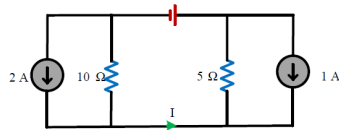


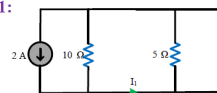
Figure 26: Superposition — worked example.

Example



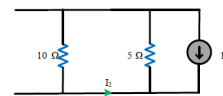
Solution:

Step 1:



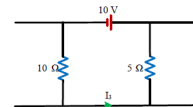
$$I_1 = 2 \frac{10}{5 + 10} = 1.33A$$

Step 2:



$$I_2 = -1 \frac{5}{5 + 10} = -0.33A$$

Step 3:



$$I_3 = - \frac{10}{5 + 10} = -0.667A$$

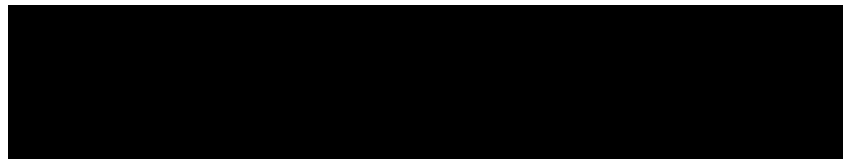
$$I = I_1 + I_2 + I_3 = 0.33A$$

Figure 27: Superposition — worked example (continued).

8.2 Thevenin's Theorem

Any linear, bilateral network having terminals A and B can be replaced by a single source of emf V_{Th} in series with a single resistance R_{Th} .

- The emf V_{Th} is the voltage obtained across terminals A and B with the load removed, i.e. the **open-circuit voltage** between A and B.
- The resistance R_{Th} is the resistance of the network measured between A and B with the load removed and the sources replaced by their internal resistances (voltage sources shorted, current sources opened).

Figure 28: Thevenin's theorem — a complex network replaced by V_{Th} and R_{Th} .

Procedure for Finding Thevenin Equivalent Circuit

1. Open the two terminals (i.e., remove any load) between which you want to find Thevenin equivalent circuit.
2. Find the open-circuit voltage between the two open terminals. It is called Thevenin voltage V_{Th} .
3. Determine the resistance between the two open terminals with all voltage sources shorted and all current sources opened. It is called Thevenin resistance R_{Th} .
4. Connect V_{Th} and R_{Th} in series to produce Thevenin equivalent circuit between the two terminals under consideration.
5. Place the load resistor removed in step (1) across the terminals of the Thevenin equivalent circuit. The load current can now be calculated using only Ohm's law and it has the same value as the load current in the original circuit.

Figure 29: Thevenin's theorem — worked example.

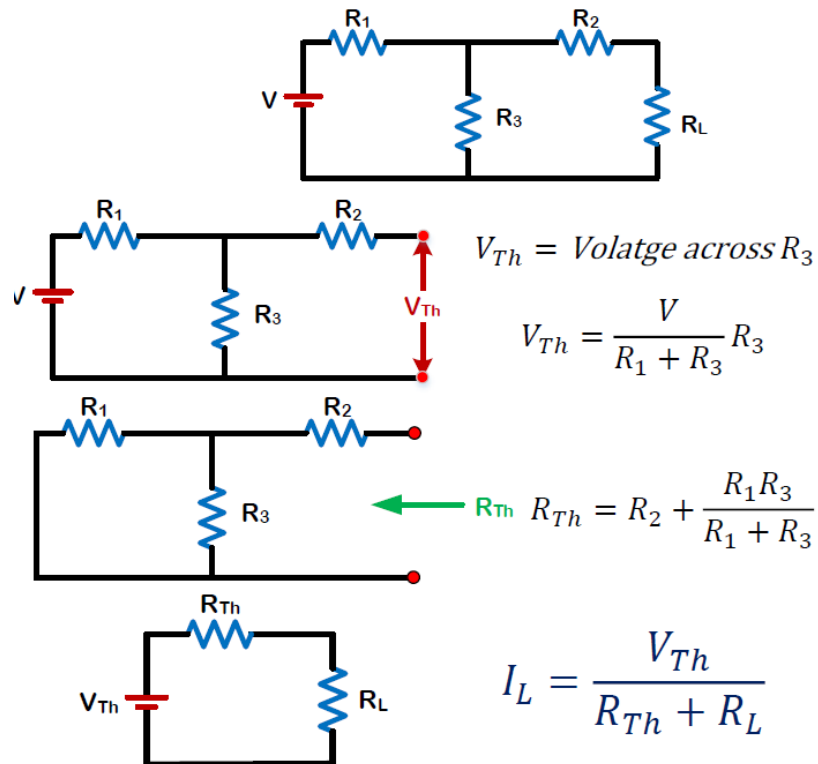


Figure 30: Thevenin's theorem — worked example (continued).

8.3 Norton's Theorem

Any linear, bilateral network having terminals A and B can be replaced by a single current source I_N in parallel with a single resistance R_N . The current I_N is the **short-circuit current** through the terminals A and B, and R_N is the resistance measured between A and B with the sources replaced by their internal resistances (the same as R_{Th}).

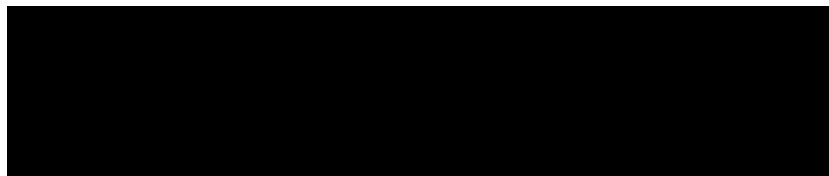


Figure 31: Norton's theorem — current source with parallel resistance.

Any two-terminal linear bilateral dc network can be replaced by an equivalent circuit consisting of a current source and a parallel resistor, as shown in Fig.

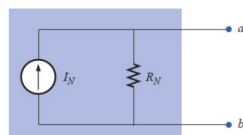


Figure 32: Norton equivalent circuit.

1. **Remove** the portion of the network across which the Norton equivalent circuit is to be found.
2. **Mark the terminals** of the remaining two-terminal network.

R_N :

3. Calculate R_N by first setting **all sources to zero** (voltage sources are replaced by short circuits, and current sources by open circuits) and then finding the **resultant resistance between the two marked terminals**. (If the internal resistance of the voltage and/or current sources is included in the original network, it must remain when the sources are set to zero.) Since $R_N = R_{Th}$, the procedure and value obtained using the approach described for Thévenin's theorem will determine the proper value of R_N .

I_N :

4. Calculate I_N by first returning all sources to their original position and finding the **short-circuit current between the marked terminals**.

Conclusion:

5. Draw the Norton equivalent circuit with the portion of the circuit previously removed **replaced between the terminals of the equivalent circuit**.

Figure 33: Norton's theorem — statement and circuit.

Relation between Thevenin and Norton equivalents: $I_N = V_{Th} / R_{Th}$ and $R_N = R_{Th}$.

8.4 Maximum Power Transfer Theorem

In DC circuits, maximum power is transferred from a source to the load when the **load resistance is made equal to the internal resistance** of the source, as viewed from the load terminals with the load removed and all emf sources replaced by their internal resistances.

Condition: $R_L = R_i$ (load resistance = internal resistance)

Maximum power: $P_{max} = V^2 / (4 R_L)$

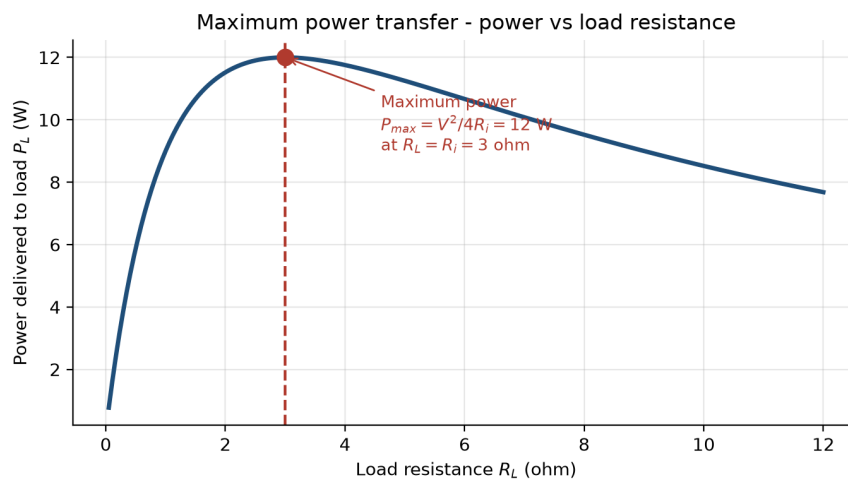


Figure 34: Power delivered to the load is maximum when $R_L = R_i$.

Important points

- The circuit efficiency at maximum power transfer is only **50%**, because one-half of the total power generated is dissipated in the internal resistance R_i of the source: $\eta = P_{out} / P_{in} = R_L / (R_L + R_i) = 1/2 = 50\%$.
- Under maximum power transfer, the load voltage is one-half of the open-circuit voltage at the load terminals.
- Value of maximum power transferred: $P_{L,max} = V^2 / (4 R_L)$.

In d.c. circuits, maximum power is transferred from a source to load when the load resistance is made equal to the internal resistance of the source as viewed from the load terminals with load removed and all e.m.f. sources replaced by their internal resistances.

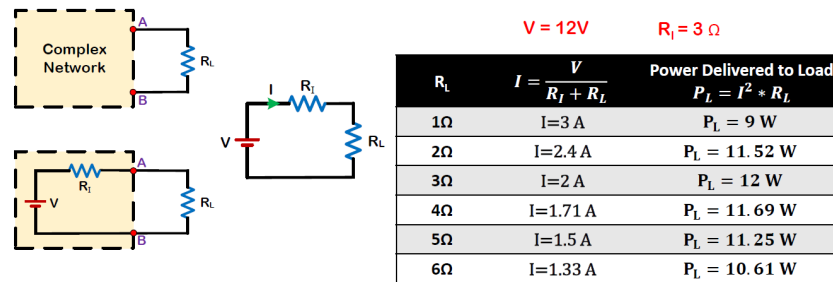


Figure 35: Maximum power transfer — worked example.

9. Solved Numerical Examples

Example 1: Thevenin's Theorem

Problem: find the current flowing through the 6 Ω resistor using Thevenin's theorem (circuit: a 4.5 V and a 3 V source with 4 Ω and 5 Ω resistors).

Solution: Thevenin voltage $V_{Th} = 3.83 \text{ V}$ and Thevenin resistance $R_{Th} = 2.22 \text{ Ω}$.
Hence the load current is:

$$I = V_{Th} / (R_{Th} + R_L) = 3.83 / (2.22 + 6) = 0.47 \text{ A}$$

Example 2: Thevenin's Theorem with a current source

Problem: find the current flowing through the 1 Ω resistor using Thevenin's theorem (12 V source, 2 A source, 6 Ω and 4 Ω resistors).

Solution: $V_{Th} = 6 \text{ V}$, $R_{Th} = 3 \text{ Ω}$. Hence:

$$I = V_{Th} / (R_{Th} + R_L) = 6 / (3 + 1) = 1.5 \text{ A}$$

Example 3: Maximum Power Transfer

Problem: find the value of resistance R for maximum power transfer in a circuit with a 120 V source, a 6 A source and 10 Ω and 5 Ω resistors, and obtain the maximum power.

Solution: $V_{Th} = 60 \text{ V}$, $R_{Th} = 10 \parallel 5 = 3.33 \text{ Ω}$. For maximum power, $R = R_{Th} = 3.33 \text{ Ω}$, and:

$$P_{max} = V_{Th}^2 / (4 R_L) = 60^2 / (4 \times 3.33) = 270 \text{ W}$$

Example 4: Series-Parallel Resistance

For a series-parallel network, first reduce each parallel group using $R_p = R_1 R_2 / (R_1 + R_2)$, then add the series resistances. For example, with a 140 Ω delta-to-star conversion, the equivalent star resistances come out as $R_a = 140 \text{ Ω}$, $R_b = 70 \text{ Ω}$ and $R_c = 35 \text{ Ω}$.

10. Quick Reference — Key Formulae and Definitions

10.1 Key Formulae

Quantity	Formula	Notes
Ohm's law	$V = I R$	V in volts, I in amperes, R in ohms
Power	$P = VI = I^2 R = V^2/R$	In watts
Energy	$W = P \times t$	1 Wh = 3600 J
Resistance of a conductor	$R = \rho l / a$	ρ in ohm-metre
Series resistance	$R_{eq} = R_1 + R_2 + \dots$	Same current through all
Parallel resistance	$1/R_{eq} = 1/R_1 + 1/R_2 + \dots$	Same voltage across all
Two resistors in parallel	$R_{eq} = R_1 R_2 / (R_1 + R_2)$	Product over sum
Voltage division	$V_1 = V R_1 / (R_1 + R_2)$	Series circuit
Current division	$I_1 = I R_2 / (R_1 + R_2)$	Parallel circuit
Source transformation	$I_S = V_S / R_S$	$R_P = R_S$
Norton from Thevenin	$I_N = V_{Th} / R_{Th}$	$R_N = R_{Th}$
Max power transfer	$R_L = R_i$; $P_{max} = V^2 / (4R_L)$	Efficiency = 50%

10.2 Key Definitions at a Glance

Term	Definition
Electric circuit	An interconnection of elements with at least one closed path in which current can flow.
Active element	An element capable of supplying energy to a circuit (voltage/current sources, generators).
Passive element	An element that only receives energy (resistor, inductor, capacitor).
Ohm's law	Current is directly proportional to voltage and inversely proportional to resistance.
KCL	The algebraic sum of currents at a junction is zero.
KVL	The algebraic sum of emfs and voltage drops around a closed loop is zero.
Node / Branch	A junction where two or more elements meet / a section between two junctions.
Loop / Mesh	Any closed path / a closed path containing no other loop.
Superposition theorem	Total response = algebraic sum of responses due to each source acting alone.
Thevenin's theorem	Any linear network = a voltage source V_{Th} in series with R_{Th} .
Norton's theorem	Any linear network = a current source I_N in parallel with R_N .
Maximum power transfer	Maximum power is delivered to the load when $R_L = R_i$.